Telco Customer Churn Prediction using Different Machine Learning Algorithms

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Abstract

Customer churn is a significant problem for companies, as losing customers directly impacts a company's revenue and growth. This project develops a machine learning model to predict customer churn using the IBM Telco Customer Churn dataset. This dataset contains a little over 7,000 customer records with features such as account information, billing behavior, and demographic information. The data preprocessing techniques that were used includes feature engineering, handling missing values, and class imbalance correction using SMOTE. These were applied to improve the models performance. The two classification algorithms, Logistic Regression and Random Forest, were implemented and evaluated using metrics like recall, accuracy, precision, and F1-score. The results conclude that both models perform well, however, the Random Forest achieved the best overall performance and recall. The analysis also identifies important factors that may influence churn, like, tenure, monthly charges, and contract type. These findings provide valuable insights that can help enhance businesses' customer retention strategies.

I. Introduction

Customer retention is a big issue for companies, especially those in competitive markets like telecommunications and financial services. Customer churn is defined as the loss of customers over time, this can significantly impact a businesses profitability and growth. Due to this, predicting and reducing churn is a very important problem in data analysis and machine learning.

Machine Learning techniques have proven quite effective in solving predictive problems via identifying patterns within large data sets. These models enable organizations to anticipate customer behavior and take proactive measures to retain customers.

In this project, we used the IBM Telco Customer Churn dataset to predict customer churn with a machine learning model. The project includes applying

classification models, handling class imbalance, feature engineering, and data preprocessing. The goal was to not only build an accurate predictive model but also to identify the key factors that influence customer churn.

II. Dataset & Data Preparation

The dataset we used is the IBM Telco Customer Churn dataset. This dataset contains approximately 7,043 customer records and over 30 features. These features include information like account information such as tenure and contract type, customer demographics, and billing-related variables such as monthly charges and total charges.

Multiple preprocessing steps were required before modeling. We removed irrelevant and leakage-prone features like CustomerID, churn related scores, and geographical information to prevent bias in the model. The feature Total Charges was converted into a numeric format, and missing values were handled using median. We then transformed categorical variables into numerical format using one-hot encoding. The dataset was then split into training and testing data sets using an 80/20 stratified split to preserve class distribution.

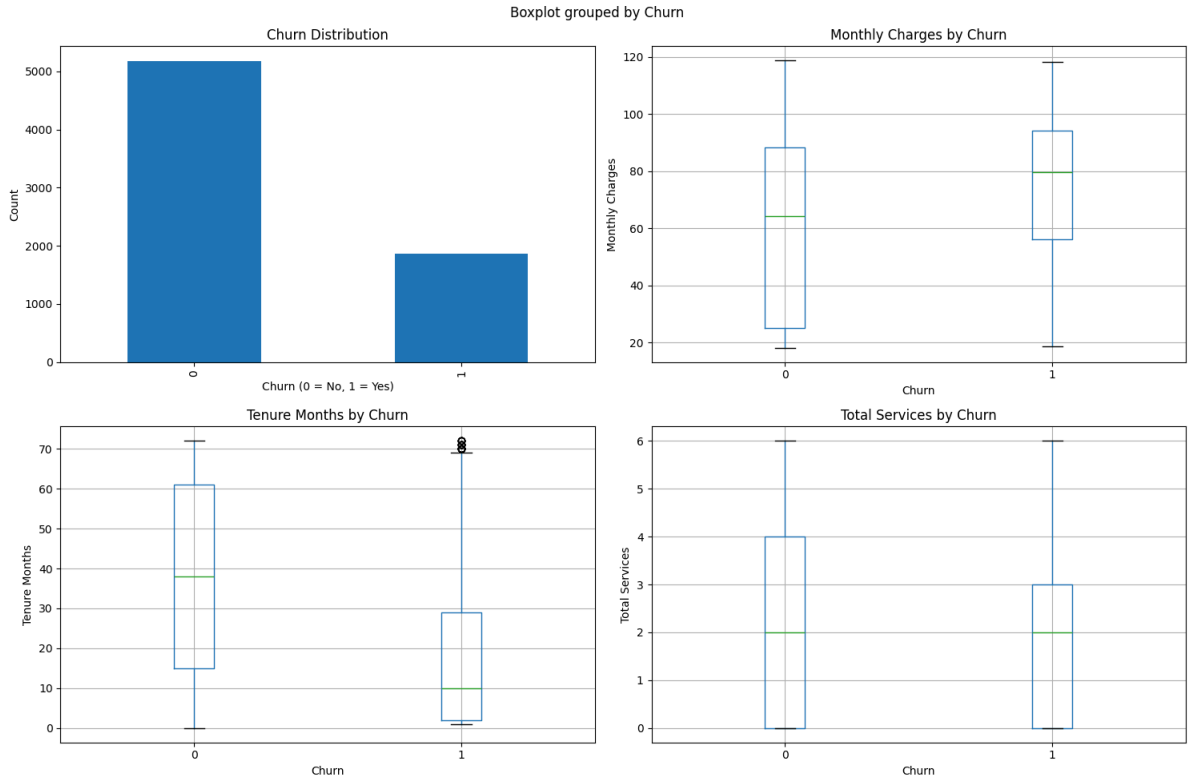
The dataset displayed class imbalance, with approximately 26.5% of customers classified as churners. We used the Synthetic Minority Oversampling Technique (SMOTE) to address this. SMOTE was then applied to the training data to help improve the model's ability to detect churn.

In addition, several new features were engineered to improve the predictive performance of the model. This includes the following: average monthly upcharge, tenure group, total number of services, and indicators for service usage.

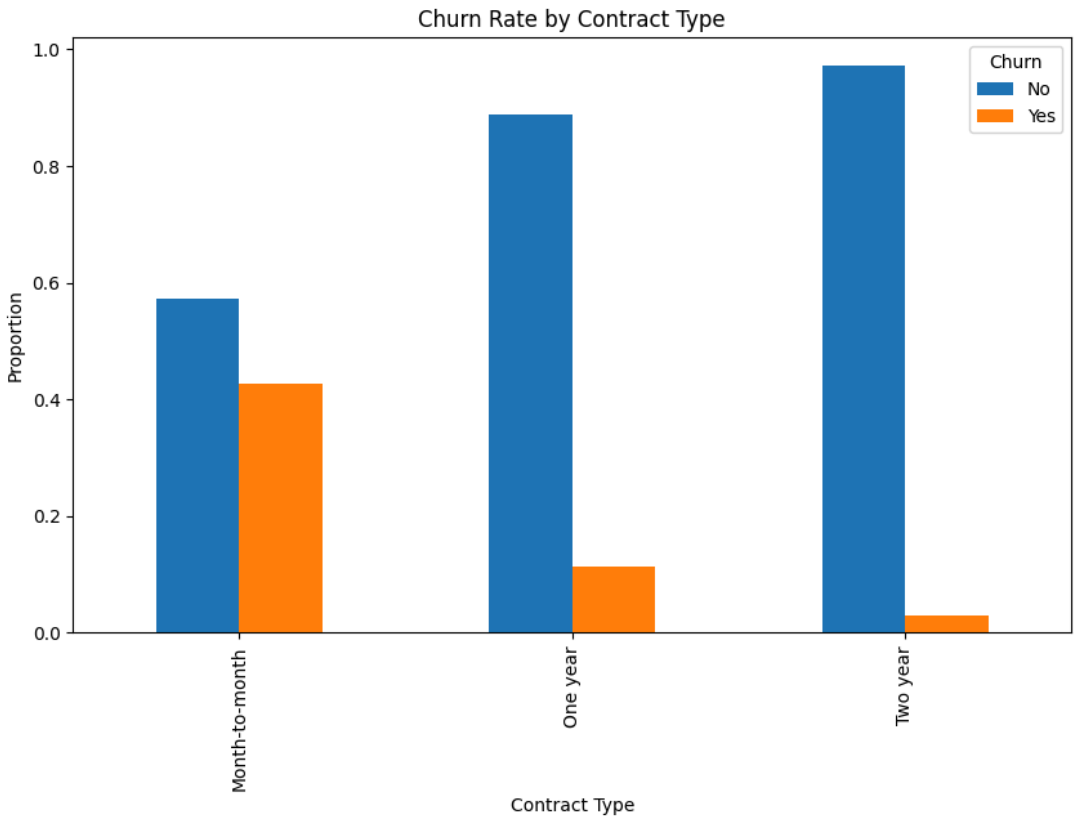
III. Exploratory Data Analysis

The exploratory data analysis focused on identifying patterns in customer behavior that could influence churn. This information was displayed as data plots that revealed clear indicators in the types of customers who churn (leave the company) and those who remain.

Based on the results of these plots, customers with shorter tenure were significantly more likely to churn, indicating that churn risk is highest early in the customer lifecycle. This suggests that retention strategies should focus more on onboarding and early engagement.



The results also indicate that customers with higher recurring costs tend to churn more frequently. This pattern suggests that customers may view the product as unaffordable or not worth their time if the price is too high.



Contract type appears to have the strongest impact of whether or not a customer will churn. Customers on monthly contracts show significantly higher churn rates compared to those on yearly or 2-year contracts. This shows that commitment level plays a significant role in customer retention and suggests that incentives for longer contracts could reduce churn.

Additionally, service-related features (such as technical support and online security) showed meaningful differences between customers who decided to churn and customers who chose to remain. Customers lacking these services were more likely to churn, suggesting that including these services may improve customer retention.

Overall, the dataset exhibits class imbalance, with fewer churn cases than non-churn cases. This imbalance is important for model development, as it can lead to biased predictions if not properly addressed.

IV. Methodology

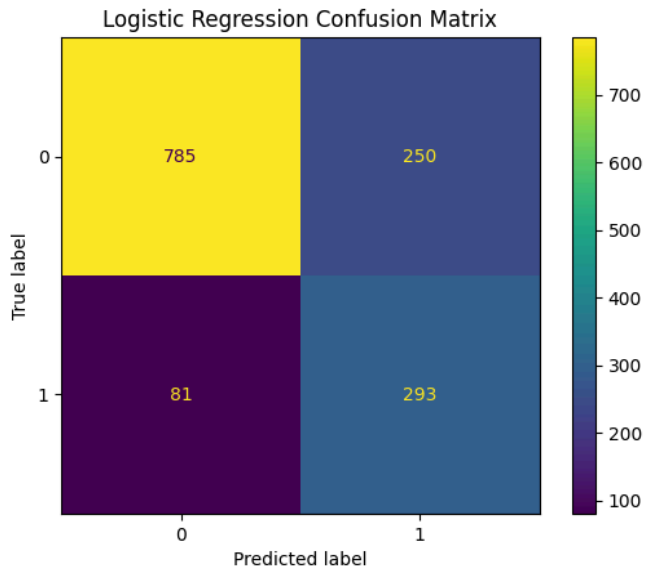
The first of the models we decided to use was a logistic regression. We chose logistic regression because it is well regarded as a simple way of determining which particular features in the dataset impact the target feature (in our case, customer churn) to the greatest degree. For our second model, we decided to use a random forest. This ensemble method should be resilient in the face of outliers and could theoretically handle a potential non-linear relationship in the data better when compared with the logistic regression.

Our target metric for this problem was recall, as opposed to accuracy, precision, or F1-score, though we kept track of each of those as well. Recall is more important than the other metrics for this particular use case because it puts emphasis on not missing customers who are about to stop doing business with the company, rather than harshly penalizing the model for believing a loyal customer is soon to churn. This reflects that the company doesn't have a significant amount to lose by offering a loyal customer a slightly better deal than is necessary, whereas the downside risk of a paying customer leaving is notable.

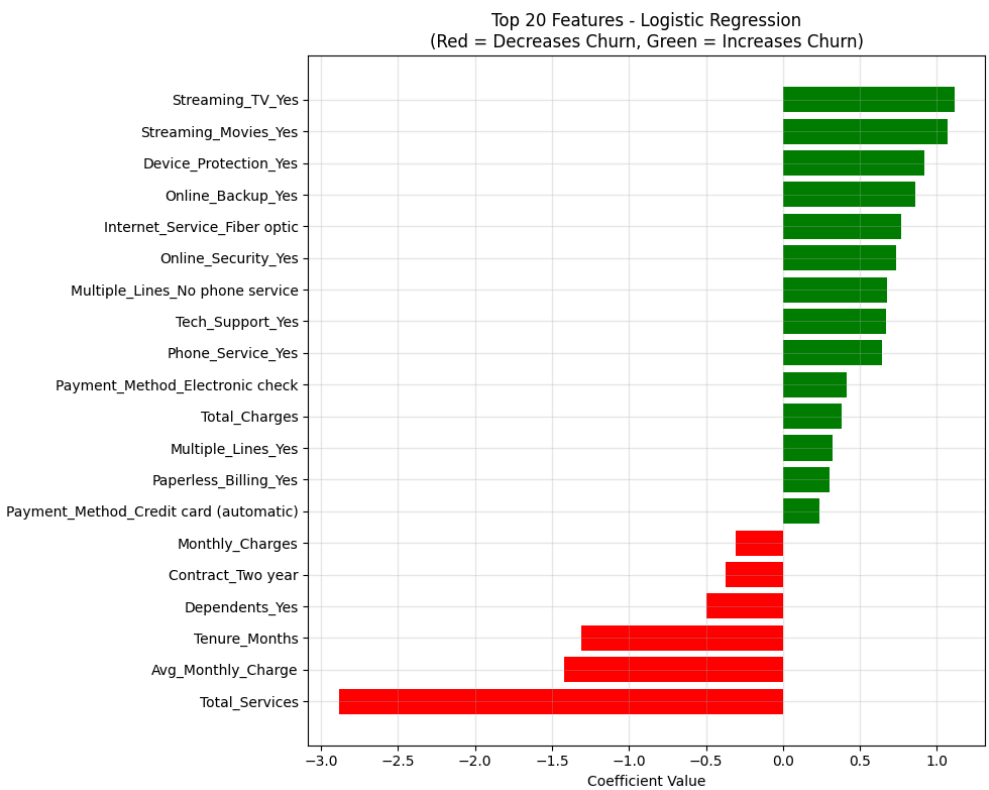
To better facilitate a high recall (at the cost of some precision) we decided to implement a custom confidence threshold of 0.35 as opposed to the typical 0.5 to determine that the customer is likely to churn. This catches more borderline cases, where the model isn't exactly sure either way, and hedges the safest bet from the perspective of the telecommunications company.

V. Results & Analysis

Our logistic regression model achieved a validation recall of 78.34% and a validation accuracy of 76.51%. While our precision and F1 score were low in comparison, at 53.96% and ~0.64 respectively, our main focus was, as noted previously, recall, due to its importance for a potential company in this type of business.

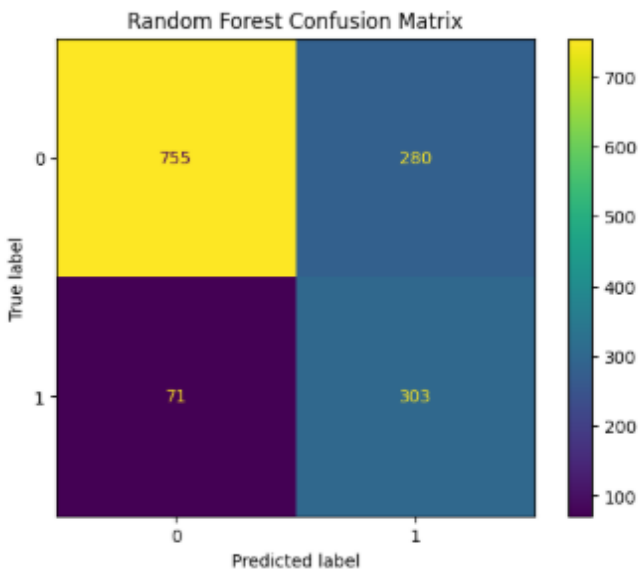


Above is a confusion matrix of our logistic regression model, showing that its prediction that a customer will not churn (or 0, in this particular graphic) is very likely to be correct, as it only predicts that when it is almost certain. Since recall aims to minimize the bottom left square and maximize the bottom right square, without regard for the top row, our model performs acceptably well.

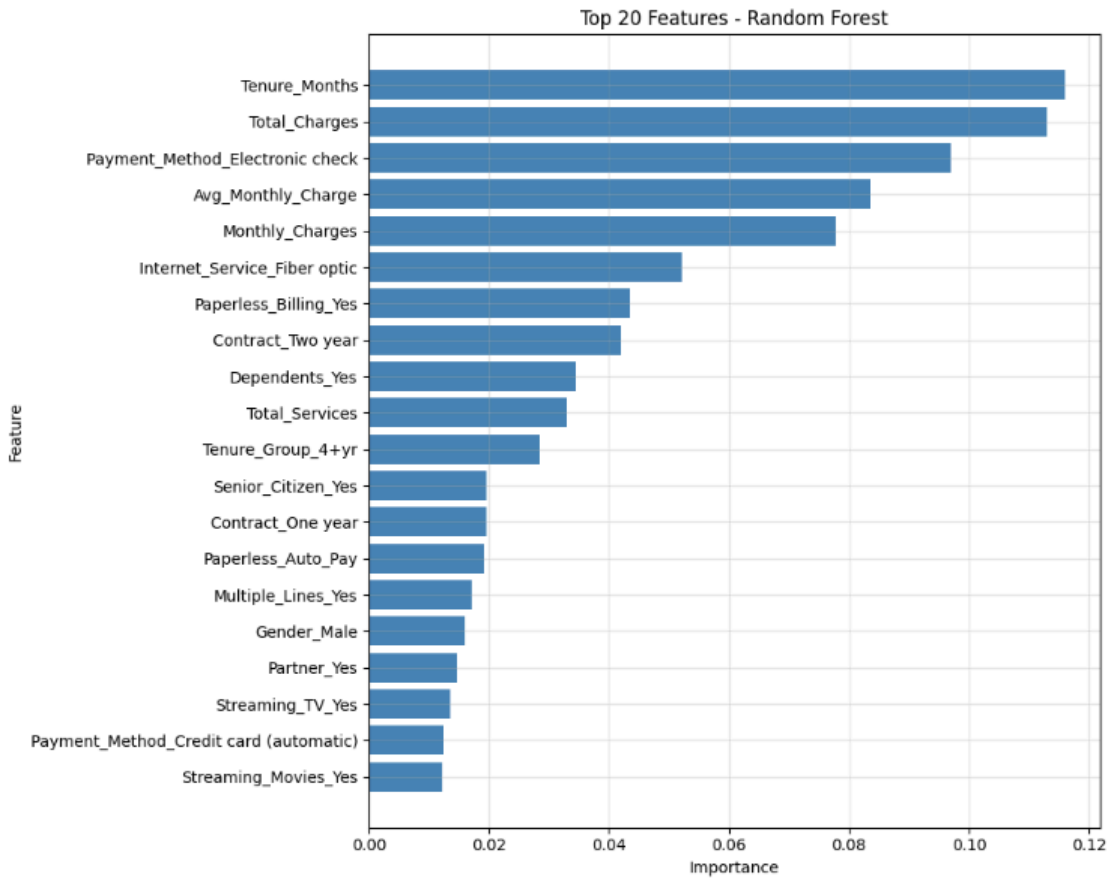


Next is a feature importance graph for our logistic regression (shown above) where we can see the most impactful features to our model's predictions. The feature

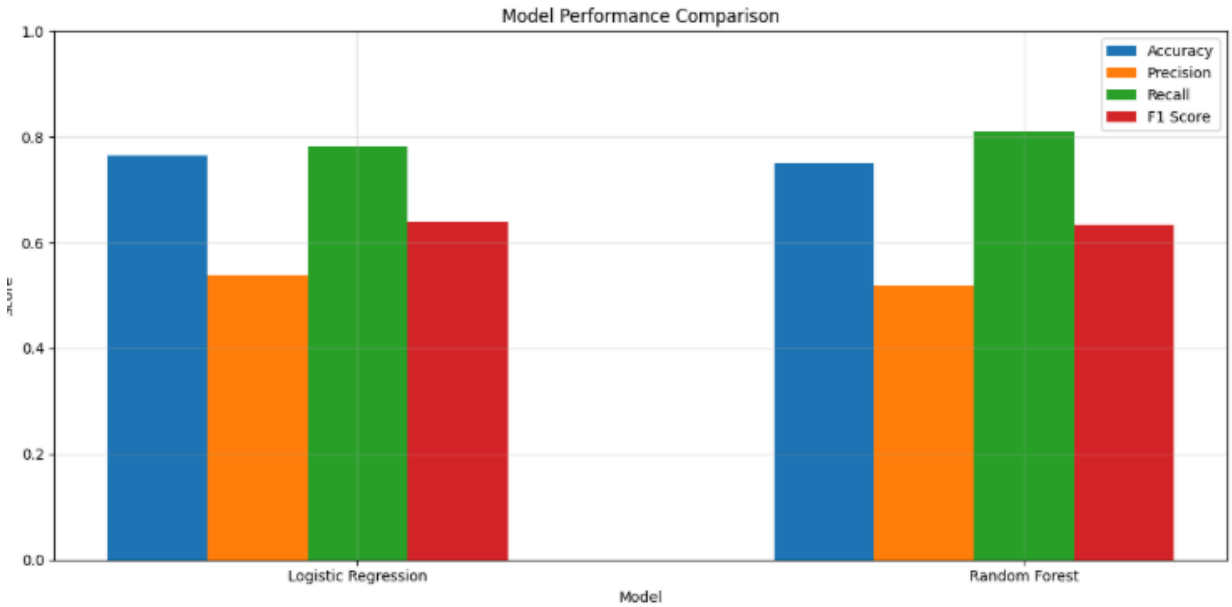
with the single highest absolute importance is Total_Services, one of our engineered features which is simply a count of how many extra services (those being 'Online_Security', 'Online_Backup', 'Device_Protection', 'Tech_Support', 'Streaming_TV', and 'Streaming_Movies') the customer pays for. A possible interpretation of this is that as the customer pays for more extra services, they are more likely to be loyal because there would be more significant friction to them switching to another service provider when compared with someone who only pays for the base service. Also nearing the top of the importance is each of these extra services' individual features, which are actually important in the opposite direction from Total_Services. This could imply that they are taking on different roles in the logistic regression; each individual service increases the cost of the plan and therefore makes the customer more likely to churn, but the total amount of services they have actually adds to their switching costs which makes them more likely to stay loyal.



Now we've come to our next model, the random forest. We saw surprisingly similar metrics in our results from this model; 75.09% accuracy, 51.97% precision, 81.02% recall, and an F1 score of ~0.63. As can be seen in the confusion matrix above, there were few differences in our metrics/guesses when compared to the logistic regression. The random forest model predicts 0 slightly less often, in both situations where the value is actually 0 and situations where it is 1, leading to the small differences in metrics we see.



The next figure is the feature importance graph for the random forest model. Interestingly, this is significantly different from the feature important graph for the logistic regression model. The random forest holds the amount of money the customer spends (Total_Charges) and the length of time they've been a customer (Tenure_Months) to be much more important for predicting customer churn, perhaps as one might expect when thinking about their own experience with telecommunication companies.



Seeing these differences in internal methodology between our two models makes it even more surprising that they perform so similarly, as can be seen in the graphical representation of their metrics' comparisons above. It seems probable that the logistic regression's representation of importance is put through some layers of abstraction (e.g. the Total_Services vs individual services) which makes it more difficult to tell exactly what the model is attributing to each feature, whereas the random forest is plotting a simpler path that follows a more human way of predicting whether a customer will churn. Due to its ease of understanding and outperformance in the all-important recall metric, we prefer the random forest model for this application.

VI. Key Insights

From the analysis there were several important key insights that were identified. Customers with short tenure are more likely to churn. Customers with higher monthly charges also have a higher probability of churn. Those with month-to-month contracts have the highest churn rates. It appears that service usage patterns also influence customer retention. This information can help businesses identify key metrics that would influence customer churn and identify high-risk customers, with this they can target retention strategies.

VII. Conclusion & Future Work

In this project we implemented a machine learning pipeline to predict customer churn. This process included feature engineering, data preprocessing, handling class imbalance, and applying classification models. Random Forest ended up being selected as the best performing model due to its strong recall and overall performance. The

project also identified important key factors that may influence a business' customer churn.

Potential future work may include testing more advanced models such as XGBoost, using larger or real time datasets, improving feature engineering, and applying deep learning techniques.

References:

IBM Telco Customer Churn Dataset,
https://community.ibm.com/community/user/businessanalytics/blogs/steven-macko/2019/07/11/telco-customer-churn-1113

https://www.kaggle.com/datasets/yeanzc/telco-customer-churn-ibm-dataset

Cole Compton - Project Contribution

For this project, I started off this project by creating the initial code for this project and the initial updated README.md file. My project code was created using jupyter notebook and included all the required functionality except it did not first import the code into mySQL. With a rating of only approximately 70%, the accuracy / precision of my project code also left much to be desired. Another member of my group corrected both issues by building on my original code. I then created an updated README.md file to describe the project so far and the work that was done, along with instructions on how to run it and a text-based diagram matching the folder structure for the project.

After creating the project code and the updated README.md file, I created a requirements.txt file and wrote instructions in the readme for setting up and running the project through a python virtual environment (venv). Lastly, I created a .gitignore file for cleaner uploads through GitHub using git commits. Lastly, I wrote the exploratory analysis section in the final project report, alongside this section for individual project contribution.

My project code first removed unnecessary and unused columns along with columns that would reveal too much data to the model and interfere with model training. Then it creates plots for visualization of the churn distribution in the imported data. It then processes the data for later use. Finally it models the data using both Logistic

Regression and Random Forest. Both models are evaluated using accuracy, precision, recall, F1-score, confusion score, and a confusion matrix. Lastly, it performs model feature importance before visualizing the data and outputting the results.

Daniel Eckman - Project Contribution

My contribution to this project began with creating a python script which initializes a MySQL database for our dataset. Then, I replaced the section of our Jupyter Notebook file where we first imported the xlsx file and changed it to import the customer churn table from our MySQL database.

After this, I noticed that there was probably room for improvement in our machine learning models and data preprocessing. First, I added the 6 new engineered features which were created from our other features we deemed to be significant. Next, I implemented SMOTE to balance our classes, since we had many more non-churners compared to churners, and I guessed that this could be a cause of some inaccuracy. Then, I set our custom threshold for each model to be 0.35 to better facilitate a high recall value.

In this document, I wrote the sections ‘Methodology’ and ‘Results & Analysis’.

Melanie Vogt - Project Contribution

My contributions to this project mainly consisted of understanding the data and developing the presentation materials along with having written most of this report. I constructed the structure of the slides, ensuring that the content was clear and aligned with the project requirements. I also drafted the sections of the final report and then wrote the heading, constructed the style, wrote the abstract, introduction, dataset and data preparation, added all of the graphs, key insights, and the conclusion and future work sections.

Throughout this project I collaborated with my group members in understanding the technical aspects of the data preprocessing and modeling process. Although my work primarily focused on the presentation and documentation I did gain important real

world understanding of how machine learning models like Logistic Regression and Random Forests are used on real datasets.

One main thing I learned from this project was the importance of communication and documentation. Even the most accurate models are not useful if they can not be clearly explained to others. Creating the presentation and report helped me grow my skills in translating technical information into understandable terms that can be better understood by a brooder audience. This project provided valuable experience in technical writing/understanding and collaborative problem solving.